ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** electron	Cathode ray experiments.
2	**(a)** protons	Z = p.
3	**(a)** 1	Gains 1 to make 8.
4	**(d)** negative	Negative charge.
5	**(d)** protons plus neutrons	A = p + n.
6	**(a)** chemically unreactive	Stable.
7	**(d)** Rutherford	Gold foil experiment.
8	**(c)** 23	11 + 12.
9	**(a)** atomic number but different mass numbers	Same Z.
10	**(d)** James Chadwick	1932.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It could not explain the scattering of alpha particles (1); It had no nucleus (1).	1+1
12	Protons = electrons = 11 (1); Neutrons = 12 (1).	1+1
13	Iodine-131 for goitre (1); Cobalt-60 for cancer treatment (1).	1+1
14	Most of the atom is empty (1); Positive charge is concentrated in a small dense nucleus (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Alpha particles directed at thin gold foil (1); Most passed; some deflected; very few rebounded (1); Nucleus small, dense and positive; mostly empty atom (1).	1+1+1
16	Isotopes: same Z, different A (1); Isobars: same A, different Z (1); Isotopes have the same chemical properties (1).	1+1+1
17	Protium: 1 proton, 0 neutron (1); Deuterium: 1 proton, 1 neutron (1); Tritium: 1 proton, 2 neutrons (1).	1+1+1
	OR: He has 2 protons but mass 4 u (1); Neutral particles of mass 1 u account for the rest (1); Helium: 2 protons + 2 neutrons (1).	1+1+1
18	Orbiting electrons should lose energy and spiral into the nucleus (1); It could not explain the stability of atoms (1); Bohr introduced fixed energy shells (1).	1+1+1
	OR: Charge = +2 (1); Magnesium ion Mg^{2+} (1); 2, 8 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Thomson: plum pudding (1); Drawback: could not explain alpha scattering (1); Rutherford: nucleus with revolving electrons (1); Drawback: instability (1); Bohr: fixed shells (1).	1+1+1+1+1
	OR: K: p 19, n 20, e 19 (1); S: p 16, n 16, e 16 (1); F: p 9, n 10, e 9 (1); Atomic number identifies the element (1); Mass number gives nucleons (1).	1+1+1+1+1
20	Thin gold foil, alpha source, fluorescent screen (1); Most passed straight: empty space (1); Some deflected: positive charge (1); Few rebounded: small dense nucleus (1); Nuclear model (1).	1+1+1+1+1
	OR: Na loses 1: 2, 8 (1); Mg loses 2: 2, 8 (1); Cl gains 1: 2, 8, 8 (1); O gains 2: 2, 8 (1); Octet attained (1).	1+1+1+1+1

Mistakes to avoid:

- Confusing atomic number with mass number, or finding neutrons as $A + Z$ instead of $A - Z$.
- Filling the M shell with 18 electrons for elements up to calcium; the outer shell cannot hold more than 8, so Ca is 2, 8, 8, 2 and not 2, 8, 10.
- Taking the valency of an element like nitrogen (2, 5) as 5 instead of $8 - 5 = 3$.
- Saying that isotopes differ in the number of protons or electrons; they differ only in the number of neutrons.
- Assuming the atomic mass of chlorine is a whole number or that every chlorine atom weighs 35.5 u; it is a weighted average of isotopes.

- Attributing the discovery of the nucleus to Thomson, or the neutron to Rutherford (nucleus: Rutherford; neutron: Chadwick; electron: Thomson).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 18	35 - 17.
2	**(d)** mass number but different atomic numbers	Same A.
3	**(d)** most of the atom is empty space	Space inside the atom.
4	**(c)** 13	Neutral atom.
5	**(b)** $^{12}_{6}\mathrm{C}$ and $^{14}_{6}\mathrm{C}$	Same Z.
6	**(c)** the nucleus is positively charged and dense	Strong repulsion near a small nucleus.
7	**(c)** 8	2n squared.
8	**(c)** $^{40}_{18}\mathrm{Ar}$ and $^{40}_{20}\mathrm{Ca}$	Same A.
9	**(a)** the nucleus is very small and dense	Rare head-on hits.
10	**(d)** 18	2 x 9.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	2, 8, 2 (1); Valency 2 (1).	1+1
12	Atoms tend to have eight electrons in the outermost shell (1); They gain, lose or share electrons to achieve this (1).	1+1
13	Electrons revolve in fixed shells without radiating energy (1); Energy is emitted or absorbed when an electron jumps between shells (1).	1+1
14	Atoms of an element with the same atomic number and different mass numbers (1); $^{35}_{17}\mathrm{Cl}$ and $^{37}_{17}\mathrm{Cl}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Na: 2, 8, 1; valency 1 (1); Al: 2, 8, 3; valency 3 (1); O: 2, 6; valency 2 (1).	1+1+1
16	They have the same number of electrons (1); Same electronic configuration (1); Chemical behaviour depends on electrons (1).	1+1+1
17	Positively charged rays (1); Discovered by Goldstein (1); Led to discovery of the proton (1).	1+1+1
	OR: H 1, He 2, Li 2,1, Be 2,2, B 2,3 (1); C 2,4, N 2,5, O 2,6, F 2,7, Ne 2,8 (1); He and Ne have complete shells (1).	1+1+1
18	Maximum 2n squared in a shell (1); Outermost shell holds at most 8 (1); Ca: 2, 8, 8, 2 (1).	1+1+1
	OR: Cobalt-60 for treating cancer (1); Iodine-131 for thyroid problems (1); Uranium-235 as a nuclear fuel (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Li: 2,1; valency 1; metal (1); F: 2,7; 1; non-metal (1); Mg: 2,8,2; 2; metal (1); S: 2,8,6; 2; non-metal (1); Ar: 2,8,8; 0; inert (1).	1+1+1+1+1
	OR: Electron: Thomson, -1 (1); Proton: Goldstein/Rutherford, +1, 1 u (1); Neutron: Chadwick, 0, 1 u (1); Mass of electron is negligible (1); Atoms are mostly empty space (1).	1+1+1+1+1
20	Hydrogen isotopes (1); Chlorine isotopes (1); Carbon isotopes (1); Same electrons so same chemical properties (1); Uses: medicine and fuel (1).	1+1+1+1+1
	OR: (10 x 20 + 11 x 80)/100 (1); = 10.8 u (1); Element is boron (1); Average mass is not a whole number (1); Because of isotopes (1).	1+1+1+1+1

Mistakes to avoid:

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- Filling the M shell with 18 electrons for elements up to calcium; the outer shell cannot hold more than 8, so Ca is 2, 8, 8, 2 and not 2, 8, 10.
- Taking the valency of an element like nitrogen (2, 5) as 5 instead of $8 - 5 = 3$.
- Saying that isotopes differ in the number of protons or electrons; they differ only in the number of neutrons.
- Assuming the atomic mass of chlorine is a whole number or that every chlorine atom weighs 35.5 u; it is a weighted average of isotopes.

- Attributing the discovery of the nucleus to Thomson, or the neutron to Rutherford (nucleus: Rutherford; neutron: Chadwick; electron: Thomson).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** iodine-131	Radioactive iodine.
2	**(c)** plum pudding model	Electrons embedded in a positive sphere.
3	**(d)** 2, 8, 1	Na.
4	**(d)** uranium-235	Fissile.
5	**(c)** fixed energy levels (shells)	Stationary orbits.
6	**(a)** 2, 8, 7	Cl.
7	**(d)** cobalt-60	Gamma rays.
8	**(b)** neutron	About 1 u each.
9	**(d)** 2	Loses 2 electrons.
10	**(d)** protium, deuterium and tritium	Same Z = 1.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Number of protons equals number of electrons (1); Positive and negative charges cancel (1).	1+1
12	Atomic number: number of protons (1); Mass number: protons plus neutrons (1).	1+1
13	Atoms of different elements with the same mass number (1); $^{40}_{18}\mathrm{Ar}$ and $^{40}_{20}\mathrm{Ca}$ (1).	1+1
14	Electron: -1, about 1/1836 u (1); Proton: +1, 1 u; neutron: 0, 1 u (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Ar: p = e = 18, n = 22 (1); Ca: p = e = 20, n = 20 (1); Isobars (1).	1+1+1
16	Combining capacity (1); Mg: 2; Cl: 1 (1); Ne: 0 as the shell is complete (1).	1+1+1
17	Phosphorus (Z = 15) (1); Valency 3 (1); Forms P^{3-} (1).	1+1+1
	OR: (35 x 3 + 37 x 1)/4 (1); = 142/4 (1); = 35.5 u (1).	1+1+1
18	Mass number = 15 + 16 (1); = 31 (1); Element is phosphorus (1).	1+1+1
	OR: Electrons in fixed shells do not radiate (1); Energy is emitted only when electrons jump (1); Stationary states (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Definition of each (1); $^{12}_{6}\mathrm{C}$ and $^{14}_{6}\mathrm{C}$ (1); $^{35}_{17}\mathrm{Cl}$ and $^{37}_{17}\mathrm{Cl}$ (1); $^{40}_{18}\mathrm{Ar}$ and $^{40}_{20}\mathrm{Ca}$ (1); Physical vs chemical properties (1).	1+1+1+1+1
	OR: Shell capacity 2n squared (1); Outermost shell has at most 8 (1); Ar: 2, 8, 8 (1); K: 2, 8, 8, 1 (1); Next shell filled only when the previous reaches its maximum limit (1).	1+1+1+1+1
20	p = e = 17, n = 18 (1); 2, 8, 7 (1); Valency 1 (1); NaCl (1); Gains 1 electron to form Cl^{-} (1).	1+1+1+1+1
	OR: Nucleus is about 10^-15 m, atom about 10^-10 m (1); Most of the atom is empty (1); Alpha particles (+) repelled by the nucleus (+) (1); Large deflection only for close approach (1); Few head-on collisions (1).	1+1+1+1+1

Mistakes to avoid:

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